

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 1: L1

Question 1 (Lecture 1)

Which of the following statements about agents are true? Select all that apply.

- (a) Human and robot agents differ because the latter do not have actuators.
- (b) Eyes and ears are examples of a human agent's actuators.
- (c) An agent senses its environment through perceptors.
- (d) Percepts are the agent's perceptual inputs from the environment at any given time.
- (e) Touchless faucets can also be viewed as agents.

Ans: (d), (e)

Question 2 (Lecture 1)

Which of the following statements regarding rational agents are true? Select all that apply.

- (a) Rational agents always know the environment dynamics and can predict future updates of the environment.
- (b) A rational agent is omniscient
- (c) A rational agent selects an action that maximizes the expected value of its performance measure.
- (d) A rational agent is not guaranteed to find optimal solutions.

Ans: (c), (d)

Question 3 (Lecture 1)

Which of the following are necessary to formally define a problem for a problem-solving agent? Select all that apply.

- (a) A clear goal condition that defines when the agent has successfully solved the problem
- (b) Specifying the outcome of each action that the agent could take

(c) The initial state from which the agent begins its problem-solving process

(d) The set of possible actions the agent can perform in each state

Ans: (a), (b), (c), (d)